

ECE 101**B. TECH. (IInd SEMESTER) EXAMINATION, 2025-26****BACHELOR OF TECHNOLOGY****(ME & CSE)****(Basic Electronics Engineering)**

Time : Three Hours]

[Maximum Marks : 75

Note: There are **three** sections (A, B and C) and candidate has to attempt questions from all sections. Marks are indicated against each section.

Section-A

1. Attempt **all** parts of the following : 5×3=15
- (a) Compare conductors, insulators and semiconductor material with the help of energy band diagram. What are n-type and p-type semiconductors ?
- (b) Differentiate between ideal and practical PN junction diode.
- (c) A JFET has a maximum drain current $I_{DSS} = 10\text{mA}$ and pinch-off voltage $V_p = -4\text{V}$. Calculate the drain current I_D when the gate-to-source voltage $V_{GS} = -2\text{V}$.

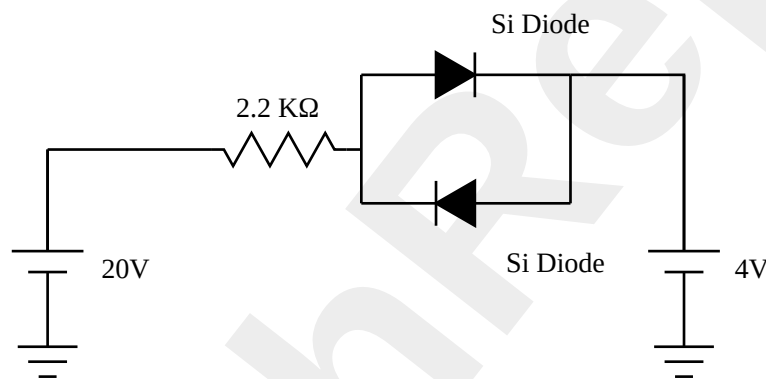
- (d) List the ideal characteristics of an Op-Amp.
- (e) State the needs of modulation in communication system.

Section-B

Note: Answer **all** questions of the following :

4×5=20

2. (a) Two diodes are connected in parallel with a $2.2\text{k}\Omega$ resistor and battery as shown in figure. Find the current I through resistor.



Or

- (b) Explain transition and different capacitance in detail.
3. (a) Discuss the common base current gain and common emitter current gain. Also derive the relationship between α , β and γ .

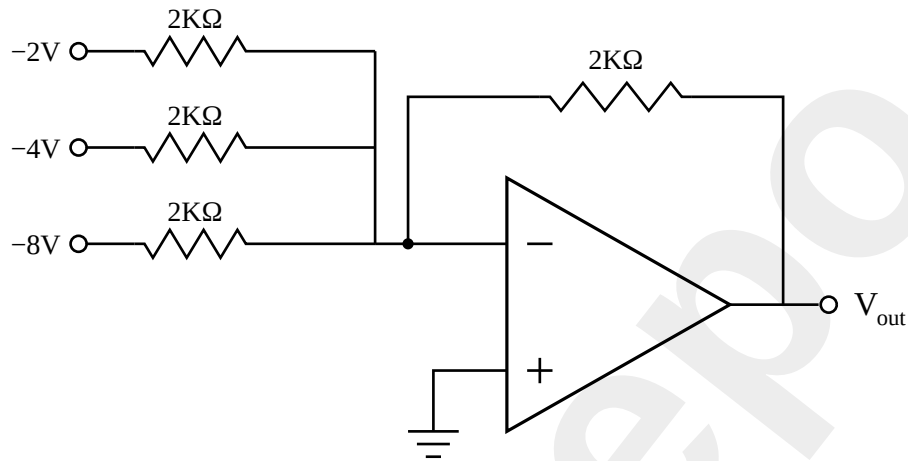
Or

- (b) Give the performance comparison of JFET and MOSFET. Why is FET a voltage controlled device ?

4. (a) Design Op-Amp Integrator circuit. Also obtain its output voltage expression.

Or

- (b) Determine the output voltage for the given circuit.



5. (a) Explain the principle of amplitude modulation.

Or

- (b) The carrier power of an Am wave is 250W. If the total power transmitted is 312.5W. Calculate the modulation index.

Section-C

Note: Answer any **two** questions of the following : $2 \times 20 = 40$

6. (a) Describe the construction, working and V-I characteristics of a PN junction diode.
- (b) Explain the working, circuit diagram and output waveform of a full-wave bridge rectifier. Also derive its V_{dc} , V_{rms} and ripple factor.

7. (a) Discuss the input and output characteristics of a BJT in the common Base Configuration.
- (b) Explain the different breakdown mechanism in PN junction diode. Explain how a Zener diode can be used as a voltage regulator with a circuit diagram.
8. (a) Describe the basic structure of an n-channel MOSFET. Also draw its transfer and drain characteristics.
- (b) Derive and explain the working of inverting and non-inverting operational amplifier.
9. (a) Explain in detail the block diagram and working of Cathode Ray oscilloscope.
- (b) (i) Explain the working of summing amplifier.
- (ii) An Am signal has a total power of 180W and a modulation index of 0.5. Find the carrier power.

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